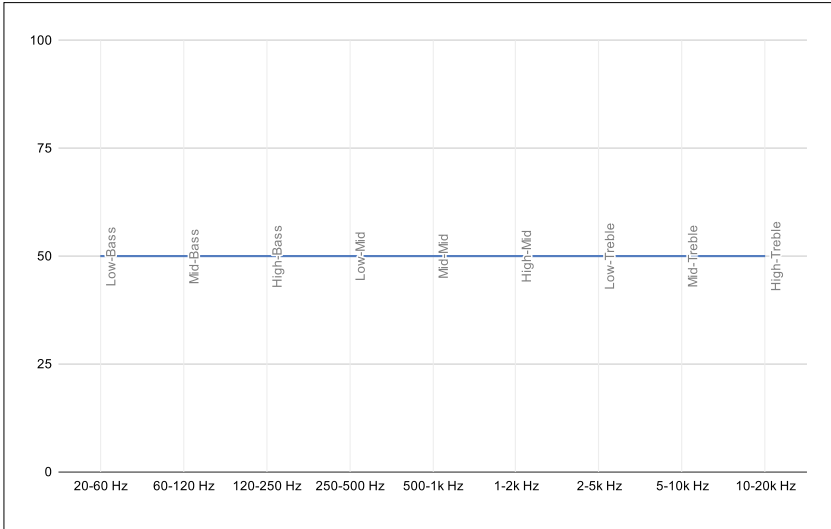




Audible frequency response graph from 20 Hz to 20 kHz

same specification will not sound the same. That is why we run multiple tests and take the help of an application and special hardware, instead of just looking at the specifications sheet. So, ensure you don't fall for the specs-race.

Yes, the size and materials of a driver will affect the way it sounds, but at the same time, the tuning will also have an impact on the overall sound signature and performance of the device.

You may have seen studio-grade headsets and regular consumer headsets from the same brands being sold and marketed differently. Right? This is because of the tuning and sound signature of those headsets. Studio-grade headphones, despite having the same drivers as consumer-grade headphones, would be tuned to have a flat response. Meaning that across frequencies, the levels would

be the same, ensuring that every element of the track or whatever is being worked on reflects as it was recorded.

Consumer-grade headphones, on the other hand, will have some colouration in their audio, meaning that the manufacturers would boost certain frequencies to make the overall listening experience more pleasurable listening experience. And, the pleasurable of the listening experience varies from consumer to consumer. So, we'd again recommend you to look out for your own preferences by trying out the headphones first.

Mind the encoder and decoder

Don't worry we did not trick you into reading our rant about programming. This is about the signal transmission

that takes place between your audio output device and your source.

Some manufacturers list the CODECs that are compatible with their devices. CODECs, in their essence, indicate the quality of the signal that can be potentially transmitted via Bluetooth to your audio device and whether or not that device would be able to convert that incoming signal into audio waves or not.

As you would have guessed, across the board, there are many types of CODECs. The most basic and common one of all is SBC. It demands the least amount of processing power from your devices and consumes less power as well. While it does come out to be the lossiest of all, its implementation by manufacturers can define the way it can perform.

Generally speaking, you should avoid going for this, but if you are using your headphones in noisy environments, without ANC, or are using a pair of headphones or earphones that are not properly tuned, then you won't even notice the difference in quality as you move higher in the CODEC spectrum. The audio signal being pushed by your streaming service, the capabilities of your smartphone all come into play when you are listening to the audio on wireless devices. In the case of wired headphones or earphones, you need to look at the capabilities of your output port.

Most mobile devices these days lack a headphone jack, so in order to get lossless audio output from your phone, you might need to get an external USB DAC specially made for mobiles. As for PCs, you will read more about buying additional

hardware for your computer in our PC components part of the cover story.

To guide you better through the world of audio CODECs, have a look at brief descriptions of some of the popular CODECs that are seen in headphones and earphones in 2023. ■

Codec	Supported Bit Rates	Special Features	Downsides	High-Res Audio Support
SBC	up to 345kbps (theoretically)	Low power consumption Low processing power required	Latency Issues Compressed audio	☐
AAC	up to 320kbps	Better compression than SBC	Not in open domain Uses more power and processing	☐
aptX	up to 384kbps	Better compression algorithm than SBC	Not supported on iOS	☒
aptX HD	up to 576kbps	Improvement over aptX Higher bit rate and bit depth than aptX	Not supported on iOS	☒
aptX LL	up to 352kbps	Low latency (about 32ms)	Not supported on smartphones	☐
LDAC	up to 990kbps	Supports High-Res Audio Better support for Android devices	Proprietary Sony technology Limited presence in the market	☒
ALAC	approx. 1411kbps	Supports lossless audio	Limited to Apple devices	☒